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Evaluate Yourself

1) By applying the knowledge of chemical classification, classify each of the following into elements, compounds or mixtures.
   (i) Sugar
   (ii) Sea water
   (iii) Distilled water
   (iv) Carbon dioxide
   (v) Copper wire
   (vi) Table salt
   (vii) Silver plate
   (viii) Naphthalene balls

1.3 Atomic and Molecular Masses

1.3.1 Atomic Masses

How much does an individual atom weigh? As atoms are too small with diameter of 10^{-10} m and weigh approximately 10^{-27} kg, it is not possible to measure their mass directly. Hence it is proposed to have relative scale based on a standard atom.

The C-12 atom is considered as standard by the IUPAC (International Union of Pure and Applied Chemistry), and it's mass is fixed as 12 amu (or) u. The amu (or) unified atomic mass is defined as one twelfth of the mass of a Carbon-12 atom in its ground state.

i.e. 1 amu (or) 1u ≈ 1.6605×10^{-27} kg.

In this scale, the relative atomic mass is defined as the ratio of the average atomic mass factor to the unified atomic mass unit.

Relative atomic mass (A_r)

$$= \frac{\text{Average mass of the atom}}{\text{Unified atomic mass}}$$

For example,

Relative atomic mass of hydrogen $(A_r)_H$

$$= \frac{\text{Average mass of H-atom (in kg)}}{1.6605 \times 10^{-27}\ \text{kg}}$$

$$= \frac{1.673 \times 10^{-27}\ \text{kg}}{1.6605 \times 10^{-27}\ \text{kg}}$$

$$= 1.0078 \approx 1.008\ \text{u}.$$

Since most of the elements consist of isotopes that differ in mass, we use average atomic mass. Average atomic mass is defined as the average of the atomic masses of all atoms in their naturally occurring isotopes. For example, chlorine consists of two naturally occurring isotopes $_{17}Cl^{35}$ and $_{17}Cl^{37}$ in the ratio 77 : 23, the average relative atomic mass of chlorine is

$$= \frac{(35 \times 77) + (37 \times 23)}{100}$$

$$= 35.46\ \text{u}$$

1.3.2 Molecular Mass

Similar to relative atomic mass, relative molecular mass is defined as the ratio of the mass of a molecule to the unified atomic mass unit. The relative molecular mass of any compound can be calculated by adding the relative atomic masses of its constituent atoms.

For example,

i) Relative molecular mass of hydrogen molecule (H_2)